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The power of critical mass to make a difference: how gender diversity in board affect US corporate carbon performance 🛒

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Purpose

This study aims to examine the impact of board gender diversity on company greenhouse gas (GHG) performance, the influence of a critical mass of women on boards on carbon performance (CP) score and its three components separately (Scope 1, Scope 2 and Scope 3). This study examines the presence of institutional investors as a contingent factor that intensifies the effectiveness of the critical mass of female directors on CP.

Design/methodology/approach

Using a sample of the US companies listed on Securities and Exchange Commission for the period 2011–2018 and making a total of 2416 observations. This study shows that reaching a critical mass of female board members enhances the level of CP. In addition, this study finds that the presence of institutional investors positively moderates this relationship.

Findings

The main results suggest that there is a nonlinear relationship between a critical mass of women directors and CP, and that institutional investors play a strategic role in shaping this relationship. The effect of institutional investors on the three components of CP is also analyzed.

Research limitations/implications

This research is characterized by the methodology adopted for a quantitative variable for measuring CP. Indeed, other research the proxies related to carbon measurements are often used as a simple binary variable. This study verifies the harmony of the

theory of critical mass measuring diversity within the board of directors, the presence of institutional investors on GHG emissions (Scope 1, Scope 2 and Scope 3), unlike previous studies (Tingbani *et al.*, 2020; [Nuber and Velte, 2021](#)) which only focus on the two measures of carbon emissions (Scope 1 and Scope 2).

Originality/value

This study shows identically that gender diversity on the board must reach a critical mass of three women directors to motivate and influence CP. We fill the gap in previous research regarding the role played by the institutional environment of the firm in improving CP. Third, this study highlights the relevance of having a critical mass of pressure-resistant female directors on boards due to their engagement in climate change issues and CP.

Keywords: [Institutional investors](#), [Board diversity](#), [GHG](#), [Carbon disclosure project](#), [Curvilinear relationship](#), [Critical mass](#), [Carbon performance](#)

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